Differential Equations

Recall from last class:

- If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v, called an **eigenvector** corresponding to λ, such that
$$Av = \lambda v.$$
- If λ is an eigenvalue of an $n \times n$ matrix A, then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ.
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \ldots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_p$ of an $n \times n$ matrix A, then $v_1, \ldots, v_p$ are linearly independent.
- If A is an $n \times n$ matrix, then the characteristic polynomial of A, denoted by $p_A(\lambda)$, is given by
$$p_A(\lambda) = \det(A - \lambda I_n).$$
- λ is an eigenvalue of an $n \times n$ matrix A if and only if λ is a zero of the characteristic polynomial.
- If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).
- An $n \times n$ matrix A is **diagonalizable** if A is similar to a diagonal matrix.
- **Diagonalization Theorem:** An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A. In this case the diagonal entires of D are the corresponding eigenvalues of A.
- **Theorem 6:** An $n \times n$ matrix A with n distinct eigenvalues is diagonalizable.
- **Theorem 7:** Let A be an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \ldots, \lambda_p$.
    1. For $1 \leq k \leq p$, the dimension of the eigenspace of λ_k is at most the algebraic multiplicity of the eigenvalue λ_k.
    2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces is n if and only if the characteristic polynomial factors into linear factors and the dimension for each eigenspace is exactly the algebraic multiplicity.
    3. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k, then $\mathcal{B}_1, \ldots, \mathcal{B}_p$ is an eigenbasis for $\mathbb{R}^n$.

1. Recall the differential equation $y''(t) = -y(t)$ as modeling a unit mass on a spring. Solve the differential equation.

2. Consider $x'(t) = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} x(t)$. Solve the differential equation.

3. Consider $x'(t) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} x(t)$. Solve the differential equation.

Differential Equations – Answers / Solutions

1. Recall we solve second order constant coefficient differential equations by guessing a solution of the form $y(t) = e^{rt}$ for r is a constant. Plugging in to the differential equation we get

$$(r^2 + 1)e^{rt} = 0,$$

and hence $r^2 + 1 = 0$. Thus we get solutions when $r = \pm i$. Thus we get two independent solutions e^{it} and e^{-it}. The general solution is then

$$y(t) = c_1 e^{it} + c_2 e^{-it}.$$

We call this the complex form of the general solution. Instead we might prefer to have a real form for the solutions. Using $e^{it} = \cos t + i \sin t$, we get

$$y(t) = c_1(\cos t + i \sin t) + c_2(\cos t - i \sin t) = (c_1 + c_2)\cos t + i(c_1 - c_2)\sin t.$$

Now notice that the real and imaginary parts of $(c_1 + c_2)\cos t + i(c_1 - c_2)\sin t$ are themselves solutions to the differential equation, and hence we get the real form of the general solution is

$$y(t) = d_1 \cos t + d_2 \sin t.$$

On the other hand we can let $v(t) = y'(t)$, then we get the system of first order equations

$$\begin{cases} v'(t) = -y(t) \\ y'(t) = v(t), \end{cases}$$

or

$$\begin{bmatrix} v(t) \\ y(t) \end{bmatrix}' = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v(t) \\ y(t) \end{bmatrix} = A \begin{bmatrix} v(t) \\ y(t) \end{bmatrix}.$$

Thus we want to solve the system

$$\vec{y}\prime(t) = A\vec{y}.$$

We guess a solution of the form $y(t) = \vec{v}e^{\lambda t}$ and we get

$$\vec{v}\lambda e^{\lambda t} = A\vec{v}e^{\lambda t},$$

or

$$(A\vec{v} - \lambda\vec{v})e^{\lambda t} = 0.$$

Thus $y(t)$ is a solution if and only if $A\vec{v} = \lambda\vec{v}$. Thus $\vec{v}$ must be an eigenvector with eigenvalue λ. Thus we get the complex form of the general solution by using the eigenvalues and eigenvectors; namely,

$$y(t) = c_1\vec{v}_1 e^{\lambda_1 t} + c_2\vec{v}_2 e^{\lambda_2 t}.$$

Plugging in the eigenvalues and eigenvectors for A we get

$$\vec{y}(t) = c_1 \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{it} + c_2 \begin{bmatrix} 1 \\ i \end{bmatrix} e^{-it},$$

and hence

$$y(t) = c_1(-i)e^{it} + c_2 i e^{-it}.$$

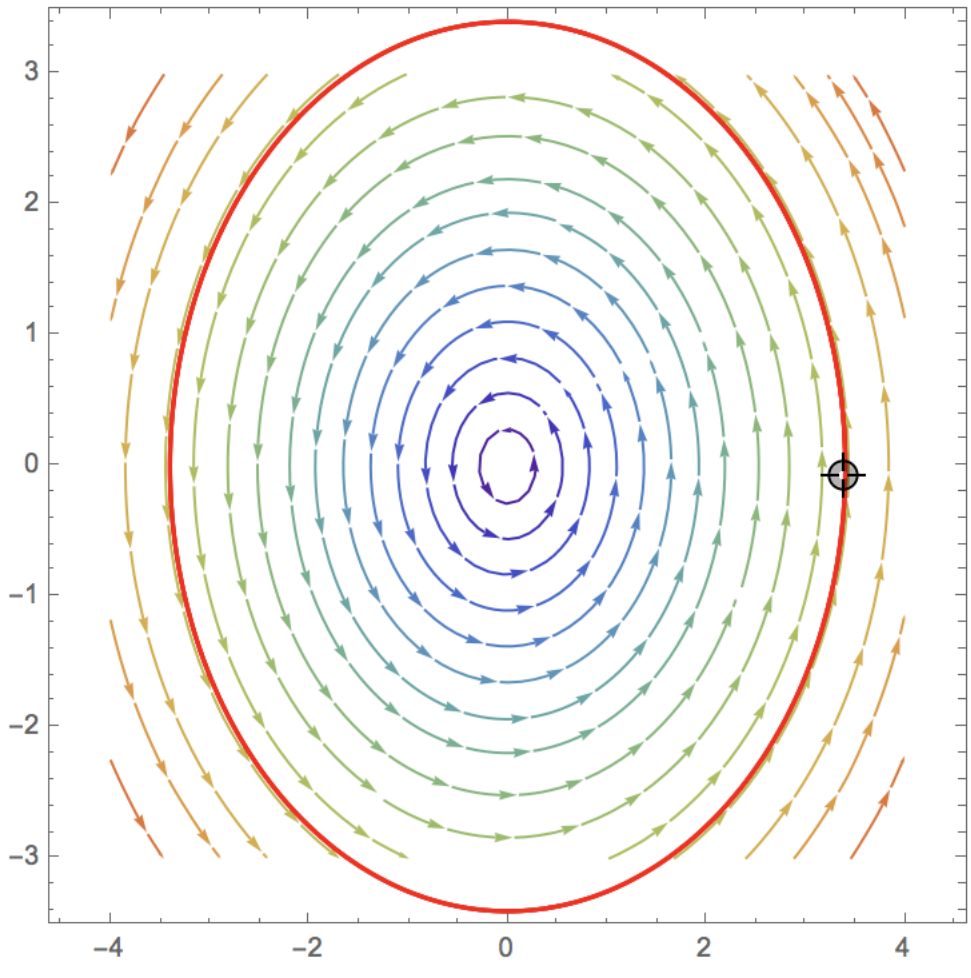
We can again use Euler's formula to get the real-form of the solution. Here we take the real and imaginary parts of $\begin{bmatrix} 1 \\ -i \end{bmatrix} e^{it}$ to get

$$\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + i \begin{bmatrix} 0 \\ -1 \end{bmatrix}\right)(\cos t + i \sin t) = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cos t - \begin{bmatrix} 0 \\ -1 \end{bmatrix} \sin t\right) + i\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \sin t + \begin{bmatrix} 0 \\ -1 \end{bmatrix} \cos t\right).$$

Again the real and imaginary parts are separately solutions to the differential equations, and hence we get the real form of the general solution is given by

$$\vec{y}(t) = d_1 \begin{bmatrix} \cos t \\ \sin t \end{bmatrix} + d_2 \begin{bmatrix} \sin t \\ -\cos t \end{bmatrix}.$$

Thus when $d_1 = 1$ and $d_2 = 0$ we get a circle of radius one in the phase portrait as shown below.



2. We again find eigenvalues and eigenvectors. Here we get eigenvalues of 4 and -1. The eigenspaces are given by

$$E_{-1} = \text{Span}\left\{\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right\},$$

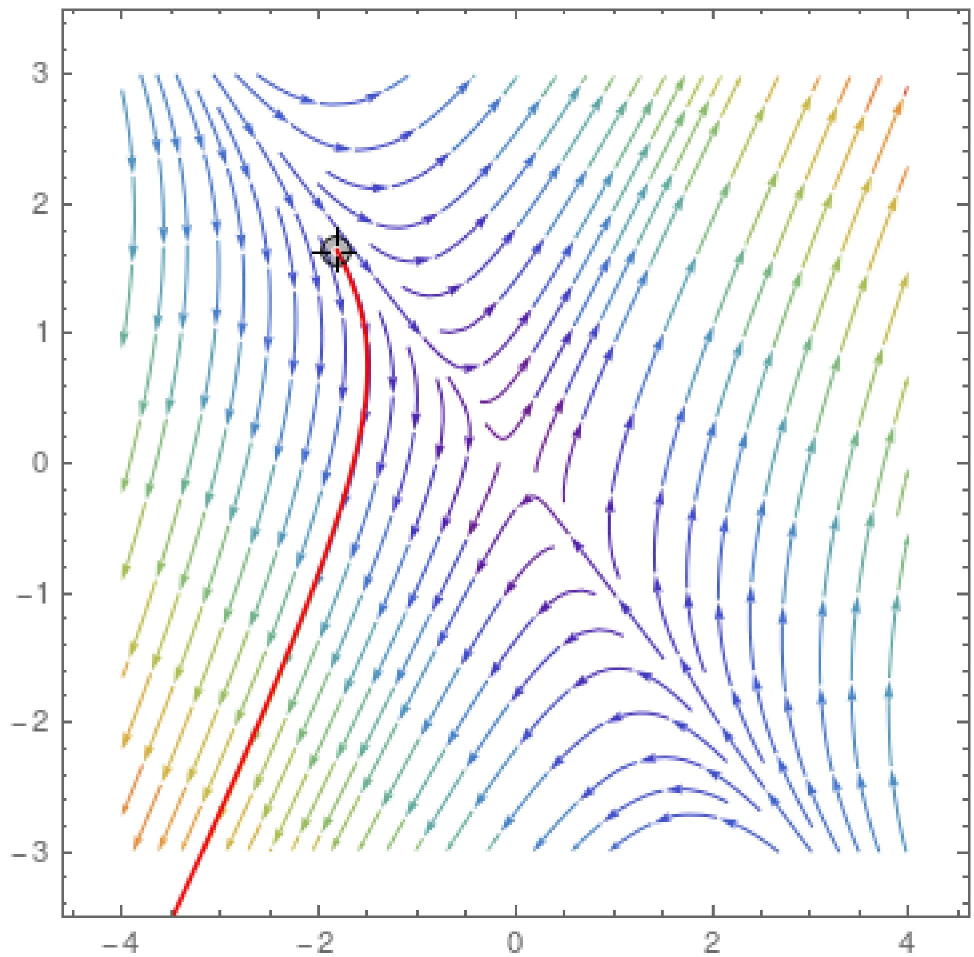
and

$$E_4 = \text{Span}\left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}.$$

Then the general solution is given by

$$\vec{x}(t) = c_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-t}.$$

We show the phase portrait with one solution trajectory in red below.



3. See the solution to question 1.